



Biofouling

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Mini-review: Inhibition of biofouling by marine microorganisms

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Any natural or artificial substratum exposed to seawater is quickly fouled by marine microorganisms and later by macrofouling species. Microfouling organisms on the surface of a substratum form heterogenic biofilms, which are composed of multiple species of heterotrophic bacteria, cyanobacteria, diatoms, protozoa and fungi. Biofilms on artificial structures create serious problems for industries worldwide, with effects including an increase in drag force and metal corrosion as well as a reduction in heat transfer efficiency. Additionally, microorganisms produce chemical compounds that may induce or inhibit settlement and growth of other fouling organisms. Since the last review by the first author on inhibition of biofouling by marine microbes in 2006, significant progress has been made in the field. Several antimicrobial, antialgal and antilarval compounds have been isolated from heterotrophic marine bacteria, cyanobacteria and fungi. Some of these compounds have multiple bioactivities. Microorganisms are able to disrupt biofilms by inhibition of bacterial signalling and production of enzymes that degrade bacterial signals and polymers. Epibiotic microorganisms associated with marine algae and invertebrates have a high antifouling (AF) potential, which can be used to solve biofouling problems in industry. However, more information about the production of AF compounds by marine microorganisms *in situ* and their mechanisms of action needs to be obtained. This review focuses on the AF activity of marine heterotrophic bacteria, cyanobacteria and fungi and covers publications from 2006 up to the end of 2012.

Keywords: marine microorganism; biofilm; inhibition; larval settlement; biofouling; antifouling defence

Introduction

In non-sterile seawater, any natural and artificial substrata quickly become fouled. Within minutes, organic molecules and particles are adsorbed onto the surface of any substratum, which is later colonised by bacteria, diatoms and protozoa (Wahl 1989) and which form a biofilm. Biofilms are composed of multiple species of microorganisms attached to the substratum (and to each other) and encased within a matrix of extracellular polymeric substances (EPS). Mature biofilms have complex, three-dimensional structures, which depend on the species composition of the biofilm, microbial activity and environmental conditions (Lewandowski 2000). Marine biofilms contain different species of heterotrophic bacteria (mainly proteobacteria), cyanobacteria, archaea and unicellular eukaryotes (eg diatoms and flagellates), while the densities of sarcodines, ciliates and fungi remain low (reviewed by Dobretsov 2010; Wahl et al. 2012). The diversity of marine organisms from biofilms has mainly been described using culture methods and microscopic observations; the number of molecular-based studies of marine biofilms remains small.

Bacteria can coordinate their adhesion, biofilm maturation, swarming, luminescence and toxin production in a process called quorum sensing (QS) (Waters & Bassler

2005; Steinberg et al. 2011). During this process, a class of small molecules named ‘autoinducers’ is produced and released into the environment. When the concentrations of these signals reach a threshold level in diffusion-limited environments, the signals trigger an expression of target genes that consequently change the behaviour of bacteria (Irie & Parsek 2008; Steinberg et al. 2011). Gram-positive and Gram-negative bacteria use a number of different, potentially overlapping QS signalling systems, but the best characterised of these is based on the production and perception of *N*-acyl homoserine lactones (AHLs) (reviewed by Dobretsov et al. 2009). Approximately 30% of the bacterial species within subtidal biofilms (Huang et al. 2009) and those associated with corals (Golberg et al. 2011) and sponges (Taylor et al. 2004) are able to produce AHLs. Some bacteria and marine organisms, such as algae, are able to respond to the QS signals of other bacteria or to compromise them (reviewed by Dobretsov et al. 2009).

Within biofilms, microorganisms interact with each other, with new microbial colonizers, as well as with larvae of invertebrates and algal spores (reviewed by Dobretsov 2010). Development of a biofilm on a substratum changes the attractiveness of the substratum to invertebrate larvae and algal spores through physical

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Table 1. Antimicrobial activity of marine heterotrophic bacteria.

Bacteria	Effective against	Source (host)	Location	Inhibitive compound	Reference
92 Isolates	Environmental and pathogenic bacteria	Nine species of red algae	India, Indian Ocean	NI	Kanagasabhapathy and Nagata (2008)
210 Isolates belonging to α -, β -, γ -proteobacteria, Firmicutes, Actinobacteria and Bacteroidetes	Pathogens	<i>Laminaria saccharina</i>	Germany, North Sea	NI	Wiese et al. (2009)
325 Isolates most belonged to α - and γ -proteobacteria	Pathogens	<i>Ulva australis</i> and <i>Delisea pulchra</i>	Australia, Pacific Ocean	NI	Penesyan et al. (2009)
<i>Pseudovibrio</i> sp. D323	Environmental bacteria	<i>D. pulchra</i>	Australia, Pacific Ocean	Tropodithietic acid	Penesyan et al. (2011)
10 Bacterial isolates	Environmental bacteria	<i>Ulva lactuca</i>	Fiji, South Pacific Ocean	NI	Kumar et al. (2011)
<i>Shewanella irciniae</i> , <i>Roseivirga spongicola</i> , <i>Gillisia myxillae</i> , <i>Fabibacter halotolarans</i> , <i>Microbulifer mediterraneus</i> , <i>Stenothermonacter spongiae</i> , <i>Winogradskyella poriferorum</i>	Environmental bacteria	<i>Lissodendoryx isodictyalis</i>	Caribbean, Atlantic Ocean	NI	Dash et al. (2009)
75 Isolates	Environmental bacteria	<i>Anoxycahyx joubini</i> and <i>Lissodendoryx nobilis</i>	Antarctica	NI	Mangano et al. (2009)
52 Isolates	Pathogens	<i>Haliclona simulans</i>	Ireland, North Sea	NI	Kennedy et al. (2009)
<i>Bacillus licheniformis</i>	Pathogens	<i>Halichondria</i> sp.	India, Indian Ocean	Indole, 3-phenylpropionic acid, 4,4'-oxybis[3-phenylpropionic acid]	Devi et al. (2010)
105 Isolates	Environmental and pathogenic bacteria	14 Invertebrates: sea urchins, brittle stars, tunicates, oysters, seaweed	Australia, Pacific Ocean	NI	Wilson et al. (2011)
<i>Pseudalteromonas haloplanktis</i>	Environmental and pathogenic bacteria	Scallop hatchery	Chili, Pacific Ocean	Isovaleric acid (3-methylbutanoic acid) and 2-methylbutyric acid (2-methylbutanoic acid)	Hayashiga-Soiza et al. (2008)

14 <i>Roseobacter</i> bacteria	Environmental bacteria	<i>Pfiesteria piscicida</i> and seawater	Denmark, North Sea	NI	Bruhn et al. (2007)
90 <i>Actinomyces</i> spp.	Pathogens	11 Sponge species	Egypt, Red Sea	NI	Abdelmohsen et al. (2010)
20 Isolates	Pathogens	9 Sponge species	Brazil, Atlantic Ocean	NI	Santos et al. (2010)
2562 Isolates	Pathogens	10 Sponge species	Germany, North Sea	NI	Muscholl-Silberhorn et al. (2008)
36 Bacteria	Pathogens	<i>Antipathes dichotoma</i>	China, South China Sea	NI	Zhang et al. (2012a)
340 Isolates	Pathogens	14 Bryozoan species	Baltic, Mediterranean Seas	NI	Heindl et al. (2010)
28 Isolates	Pathogens	<i>Balanus amphitrite</i>	India, Indian Ocean	NI	Jebasingh and Murugan (2011)
78 Isolates	Pathogens	9 Species of corals	Israel, Red Sea	Evidence of cell bound proteinaceous compounds	Shnit-Orland and Kushmaro (2009)
1138 Bacteria	Pathogens	5 Species of corals	Taiwan, South China Sea	NI	Chen et al. (2012)
519 Bacteria	Pathogens	Seawater and surface of organisms	Global waters	Tropodithietic acid	Gram et al. (2010)
101 Strains of <i>Pseudoalteromonas</i>	Pathogens	Seawater and surface of organisms	Global waters	NI	Vynne et al. (2011)
110 Bacteria	Environmental bacteria	Seawater	Denmark, North Sea	NI	Bernbom et al. (2011)
<i>Shewanella oneidensis</i> SCH0402	Environmental bacteria	Seawater	Korea, South Sea	Cis-9-oleic acid	Bhattarai et al. (2007)
<i>Pseudoalteromonas</i> sp. 3J6	Environmental bacteria and pathogens	Seawater	France, Morbihan Gulf	NI	Dheilly et al. (2010)
<i>Bacillus mojavensis</i> , <i>B. firmus</i>	Environmental bacteria	Turtle grass and limestone	Mexico, Atlantic Ocean	Surfactin A, mycosubtilin, bacillomycin D	Ortega-Morales et al. (2008)
<i>Pseudomonas rhizosphaerae</i>	Environmental bacteria	Deep sea sediments	Pacific Ocean	Cyclo(Tyr-Pro), cyclo(Tyr-Ile), cyclo(Phe-Pro), cyclo(Val-Pro) 3-phenyl-2-propenoic acid, uracil acid	Qi, Xu, Gao et al. (2009)

Note: NI = compound has not been identified.

modification of surfaces and production and release of chemical compounds (reviewed by Qian et al. 2007; Hadfield 2010; Harder et al. 2012; Mieszkin et al. 2012).

While being of relatively small size, microbial biofilms are extremely important in industry, as they can increase drag force and reduce heat transfer efficiency (Yebra et al. 2004). Biofilms can lead to corrosion of metals by increasing the open circuit potential toward positive values by metal deposition at the cathode, oxygen removal and acid production (Videla & Herrera 2005; Landousli et al. 2011). While biocidal and fouling-release coatings are generally free from macrofouling, they are still subject to colonisation by biofilms (Cassé & Swain 2006; Molino et al. 2009; Dobretsov & Thomason 2011; Zargiel et al. 2011; Briand et al. 2012). Biofilms on antifouling (AF) coatings increase roughness and hence drag, which results in increased fuel consumption of vessels (Yebra et al. 2006; Schultz et al. 2011). Thus, eradication of biofilms and inhibition of their growth are major industrial concerns.

The recent ban on toxic AF biocides (eg triorganotins) has underlined the need for the development of 'environmentally friendly' AF strategies and stimulated an active search for non-toxic natural marine AF compounds (reviewed by Qian et al. 2010). Most of these compounds have been isolated from eukaryotic organisms (Qian et al. 2010), with the disadvantage that such compounds are in limited supply. In contrast, marine microorganisms are providing a number of benefits for industries, which include the possibility of having an unlimited supply of compounds through fermenter cultivation, and the possibility of genetic and chemical modification of the source organisms and compounds, respectively (Dobretsov et al. 2006). Indeed, chemicals produced by marine microbes have been shown to be effective inhibitors of biofouling and their production to be suitable for scaling up in industrial applications (Qian et al. 2012).

This review covers the literature (including recent patents) on the inhibition of biofouling by marine bacteria, cyanobacteria and fungi from 2006 up to the end of 2012. Over this period of time, several comprehensive reviews have been published on the properties of AF compounds (Qian et al. 2010; Fusetani 2011), larval settlement (Hadfield 2010; Thiagarajan 2010), properties of epibiotic bacteria (Egan et al. 2008, 2012; Penesyan et al. 2010; Harder et al. 2012; Wahl et al. 2012) and chemical ecology (Paul & Ritson-Williams 2008; Paul et al. 2011). Interference with microbial QS as a mechanism to control marine biofilms has been the subject of another recent review (Dobretsov et al. 2009); therefore, this topic has been excluded from this article. This review is organised by the bioactivity of the microorganisms, focusing on antibacterial, antialgal and antilarval activities and on microbial disruption of biofilms.

These considerations are followed by concluding remarks.

Antibacterial activity

Heterotrophic bacteria

Bacteria with antimicrobial properties have been isolated frequently from surfaces of marine invertebrates and algae (Table 1). It is beneficial for the host to harbour epibiotic bacteria with AF properties because they provide protection against fouling. Although microbial epibiotic communities associated with marine organisms have been detected in many studies (reviewed by Wahl et al. 2012), it is far from clear whether hosts select the bacteria with AF properties, or whether such activities are simply coincidental.

Since the discovery of the first antimicrobial compound from the symbiotic bacterium *Pseudomonas tunicata* associated with the tunicate *Ciona intestinalis* (James et al. 1996), the number of studies on the antibacterial activity of epibiotic marine bacteria has increased substantially (Table 1). Of the 210 bacterial isolates from the brown alga *Laminaria saccharina*, 103 inhibited the growth of at least 1 of the pathogens tested (Wiese et al. 2009). By contrast, out of 92 isolates from Japanese red algae, only *ca* one-third of the isolates inhibited the growth of environmental and pathogenic bacteria (Kanagasabhapathy & Nagata 2008). Similarly, a low percentage (~12%) of microbes with antibacterial properties belonging to the α - and γ -proteobacteria were isolated from the green alga *Ulva australis* and the red alga *Delisea pulchra*, respectively (Penesyan et al. 2009). All these studies suggest that the proportion of antibacterial epibiotic bacteria is different for different species of algae and may vary with different geographical locations. The latter assumption is supported by the study of Heindl et al. (2010), which showed that the presence of epibiotic, cultivable bacteria with antibacterial activity was site-specific, but not species-specific.

Bacteria associated with sponges, corals, echinoderms and bryozoa have also been demonstrated to have antibacterial activity (Table 1). Four out of 7 bacteria isolated from sponges from the Caribbean inhibited the growth of bacteria isolated from Hong Kong-derived biofilms (Dash et al. 2009). Antagonistic interactions among epibiotic bacteria isolated from the sponges *Anoxycalyx joubini* and *Lissodendoryx nobilis* have been demonstrated (Mangano et al. 2009), and the microbial isolates associated with these sponges have been characterised. More than 50% of the isolates from the sponge *Haliclona simulans* inhibited the growth of pathogenic bacteria and yeast (Kennedy et al. 2009). Screening of actinomycete isolates for the presence of genes encoding polyketide synthase and non-ribosomal peptide synthase (typically involved in the synthesis of antibiotics),

revealed that these genes were common in active isolates. In contrast, another study with a high number of bacterial isolates (2562) from 10 Baltic sponge species showed that only a few of them had inhibitory properties (Muscholl-Silberhorn et al. 2008). One of the bioactive isolates related to *Pseudovibrio denitrificans* was present at high densities in all of the sponges investigated, and hence it was speculated that this strain might be responsible for AF protection of these sponge species. Four out of 28 isolates from the shells of the barnacle *Balanus amphitrite* inhibited growth of 10 human pathogens (Jebasingh & Murugan 2011). Seventy percent of the bacterial strains isolated from massive and solitary corals had antibacterial activity, while only a few isolates from branching and soft corals had such activity (Shnit-Orland & Kushmaro 2009).

Some bacterial strains isolated from seawater inhibited the growth of marine and human pathogens (Table 1; Gram et al. 2010; Bernbom et al. 2011; Vynne et al. 2011). The highest antibacterial activity was observed among *Pseudoalteromonas* and *Ruegeria* isolates. The attachment of *Pseudoalteromonas* S91 was prevented by biofilms of *Pseudoalteromonas piscicida*, *P. tunicata* and *P. ulvae* (Bernbom et al. 2011). Liquid culture of *Pseudoalteromonas* sp. 3J6 isolated from French waters inhibited the formation of a biofilm and reduced the number of viable cells of *Paracoccus* sp. 4M6, *Vibrio* sp. D01 and the pathogens *Pseudomonas aeruginosa*, *Salmonella enterica* and *Escherichia coli* (Dheilly et al. 2010).

Only a small number of natural products have been isolated from epibiotic bacteria (Table 1). *Pseudovibrio* sp. D323, isolated from the red alga *D. pulchra*, produced tropodithietic acid, which inhibited the growth of other marine bacteria (Penesyan et al. 2011). The authors speculated that tropodithietic acid prevents growth of competitors and inhibits growth of pathogens on the surface of the alga. This hypothesis was supported by the observation of Webster et al. (2008), who noticed that the loss of species producing tropodithietic acid coincided with heavy microbial colonisation of the sponge *Rhopaloeides odorabile*. Interestingly, tropodithietic acid was identified as an antimicrobial compound and was isolated from *Ruegeria* strains obtained from all ocean areas except the Arctic and the Antarctic waters during the Danish marine research expedition Galathea 3 (Gram et al. 2010). Three antibacterial compounds were isolated from methanol extracts of the bacterium *Bacillus licheniformis* associated with the sponge *Hali-chondria* sp. (Table 1; Devi et al. 2010). One of these compounds, indole, had significant activity against bacterial pathogens, while others, 3-phenylpropionic acid and 4,4'-oxybis[3-phenylpropionic acid], had moderate and weak antibacterial activity, respectively. Lipopeptides (sufracin A, mycosubtilin and bacillomycin D) produced by *Bacillus mojavensis* and *B. firmus* associated with biotic

or abiotic surfaces inhibited the growth of the bacterium *Halomonas marina*, and were toxic to the crustacean *Artemia salina* (Ortega-Morales et al. 2008). Several diketopiperazines (DKPs), uracil and 3-phenyl-2-propenoic acid produced by the deep sea bacterium *Pseudomonas rhizosphaerae* inhibited the growth of fouling bacteria (Qi, Xu, Gao et al. 2009). Isnansetyo and Kamei (2009) reviewed several of the antibacterial substances from marine *Pseudomonas* species, including pyroles, quinoline and quinolone.

The data presented in Table 1 suggest that the majority of bacterial strains isolated from marine organisms demonstrate some antimicrobial activity. The production of antibacterial compounds by bacteria is not constitutive and is not attributed to a specific taxonomic group of bacteria or specific geographical location. The specific culture conditions and currently unknown host cues affect the production of antimicrobial compounds by marine isolates cultured under laboratory conditions (Bruhn et al. 2007). For example, a strain of *B. licheniformis* lost its bioactivity when it was cultured in a liquid medium instead of on agar surfaces (Yan & Boyd 2002; Matz et al. 2008). Similarly, *Pseudoalteromonas* isolates attached to biotic or abiotic surfaces are more likely to have antibacterial properties than planktonic isolates (Vynne et al. 2011). Mucus of the coral *Acropora palmata* inhibited antibiotic and pigment production by the 'visitor' isolate *Pseudoalteromonas* sp. (Ritchie 2006). This provides evidence that marine hosts have the ability to manipulate the secondary metabolism and colonisation behaviours in the microorganisms they encounter.

Cyanobacteria

Marine cyanobacteria (blue-green algae) are a diverse group of unicellular microorganisms that exhibit a range of morphologies and have the ability to perform a range of metabolic processes such as oxygenic and anoxygenic photosynthesis, fermentation and nitrogen fixation (Stahl 1995). Both planktonic and benthic species are known, and the latter are actively involved in the formation of biofilms and microbial mats. Cyanobacteria produce chemicals in order to facilitate aggregation to form biofilms and to compete with other organisms for space and nutrients. Therefore, marine cyanobacteria have been recognised as a rich source of secondary metabolites with AF properties (Burja et al. 2001; Pulz & Gross 2004; Abed et al. 2009; Dobretsov et al. 2010; Kwan et al. 2011). Although there are more than 300 nitrogen-containing secondary metabolites with different biological activities reported from marine cyanobacteria, only a few compounds are known for their AF properties (Tan & Goh 2009). Most of these compounds, which are produced by 15 different cyanobacterial genera, have been chemically identified as lipopeptides, polyketides, amides, alkaloids, indoles and fatty acids (reviewed by Dahms et al. 2006).

Table 2. Antimicrobial compounds from cyanobacteria.

Species of cyanobacteria	Effective against	Source (host)	Location	Inhibitive compound	Reference
Family Oscillatoriaceae					
<i>Lyngbya majuscula</i>	<i>Mycobacterium tuberculosis</i>	Seawater	Guam, Pacific Ocean	Depsipeptides (ie pitipeptolides A–F)	Montaser et al. (2011)
<i>L. majuscula</i>	<i>Escherichia coli</i> , <i>Bacillus subtilis</i> , <i>B. cereus</i> , <i>Staphylococcus epidermis</i>	Culture collection	India	Methanol extract	Kaushik et al. (2009)
Family Nostocaceae					
<i>Anabaena constricta</i>	<i>B. cereus</i> , <i>Arthrospira laxissima</i> , <i>Nostoc carneum</i> , <i>Chroococcus minutus</i> , and <i>Synechosystis aquatilis</i>	Seawater	Kuwait, Persian Gulf	6-Bromo-3hydroxy-methyl-indole-2-one	Volk et al. (2009)
<i>Nodularia harveyana</i> and <i>Nostoc insulare</i>	<i>E. coli</i> , <i>P. aeruginosa</i> , <i>S. aureus</i> and <i>B. subtilis</i> , <i>Arthrospira laxissima</i>	Seawater	Germany, Baltic Sea	Norharmane (9H-pyrido(3,4-b)indole) and 4,4'-dihydroxybiphenyl	Volk and Furkert (2006)
Family Chroococcaceae					
<i>Synechocystis</i> spp., <i>Synechococcus</i> spp.	Gram-positive bacteria	Rocky shores	Portugal, Atlantic Ocean	Dichloromethane extracts	Martins et al. (2008)

The marine cyanobacterium *Lyngbya majuscula* has been shown to be a rich source of bioactive compounds including antibacterials (Table 2). Several cyclic depsipeptides (eg pitipeptolides A–F) were purified from this cyanobacterium and were shown to inhibit the growth of *Mycobacterium tuberculosis* besides having other cytotoxic activities (Montaser et al. 2011). In another study, extracts from the same cyanobacterium were shown to inhibit the growth of *E. coli*, *Bacillus subtilis*, *B. cereus*, *Staphylococcus epidermis* and *Enterococcus faecalis* (Kaushik et al. 2009). A methanolic extract of *L. majuscula* had significant activity against *B. subtilis* with a MIC value of 512 µg ml⁻¹. A new antibacterial brominated indole alkaloid, identified as 6-bromo-3hydroxymethyl-indole-2-one, has been isolated and purified from the marine cyanobacterium *Anabaena constricta* (Volk et al. 2009). Similarly, 2 new metabolites with antibacterial activity have been isolated from the marine cyanobacteria *Nodularia harveyana* and *Nostoc insulare*. The two compounds were identified as norharmane (9H-pyrido(3,4-b)indole) and 4,4'-dihydroxybiphenyl, and both inhibited the growth of *E. coli*, *P. aeruginosa*, *S. aureus* and *B. subtilis* with a MIC between 16 and 160 µg ml⁻¹.

Twenty strains of cyanobacteria isolated from hard corals infected with black band disease (BBD) and other marine sources have been tested against heterotrophic bacteria isolated from the surface mucopolysaccharide layer of healthy and infected corals (Gantar et al. 2011). Fifteen out of the 20 isolates exhibited antibacterial activity against at least 1 of the bacterial strains tested and inhibition was higher for BBD cyanobacteria. *Synechocystis* spp. and *Synechococcus* spp. had antibacterial activity against Gram-positive bacteria, but not against Gram-negative ones (Martins et al. 2008). Similarly, cyanobacteria isolated from benthic mats of Antarctic lakes have been shown to have antimicrobial activity (Biondi et al. 2008). Seventeen cyanobacteria out of the 48 isolates inhibited the growth of the bacterium *S. aureus*, the filamentous fungus *Aspergillus fumigatus* and the yeast *Cryptococcus neoformans*. The antimicrobial activity of the cyanobacterium *Geitlerine-ma* strain Flo1 was found to depend on the growth conditions (Caicedo et al. 2011).

Fungi

Many marine fungal isolates demonstrate promising antibacterial activity against pathogens (Table 3). In a recent study, more than half of the 24 fungal strains belonging to the orders Eurotiales, Hypocreales, Pleosporales and Botryosphaeriales associated with the black coral *Antipathes dichotoma*, exhibited antimicrobial activity against bacterial and fungal pathogens (Zhang et al. 2012a). Eighteen fungal isolates belonging to the Ascomycota from the gorgonian coral *Echinogorgia*

Table 3. Antifouling activity of marine fungi.

Fungal strain	Effective against	Source (host)	Location (country)	Inhibitive compound	Reference
Eurotiales Hypocreales Pleosporales Botryosphaeriales	Pathogenic microbes	<i>Antipathes dichotoma</i>	China	NI	Zhang et al. (2012a)
18 Fungal isolates	Pathogenic microbes	<i>Echinogorgia rebekka</i>	China	NI	Wang et al. (2011)
121 Isolates belonging to 20 genera	Environmental and pathogenic bacteria	6 Gorgonian corals	China	NI	Zhang et al. (2012b)
<i>Aspergillus</i> sp.	Environmental bacterial strains	<i>Xestospongia testudinaria</i>	China	Aspergiterpenoid A, (-)-sydonol, (-)-sydonic acid, Li et al. (2012) and (-)-5-(hydroxymethyl)-2-(2',6',6'-trimethyltetrahydro-2H-pyran-2-yl)phenol	
<i>Cladosporium</i> sp. F14	Environmental bacteria and <i>Bugula neritina</i>	Seawater	China	3-phenyl-2-propenoic acid, cyclo(Phe-Pro), cyclo (Val-Pro), bis(2-ethylhexyl)phthalate	Xiong et al. (2009); Qi, Xu, Xiong et al. (2009)
<i>Ampeomyces</i> sp.	Environmental bacteria, <i>Balanus amphitrite</i> and <i>Hydroides elegans</i>	Seawater	China	3-chloro-2,5-dihydroxybenzyl alcohol	Kwong et al. (2006)
<i>Cochliobolus lunatus</i>	The <i>B. amphitrite</i>	<i>Dichotella gemmacea</i>	China	14 New resorcylic acid lactones	Shao, Wang (2011)
<i>Aspergillus</i> sp.	<i>B. amphitrite</i>	<i>D. gemmacea</i>	China	Aspergilones A	Shao, Wu (2011)
<i>Penicillium</i> sp. OUCMDZ-776	<i>Bugula neritina</i>	Hypersaline sediments	China	Penispirolloid A	He et al. (2012)

Note: Fungal strains are classified by their activity against bacteria and invertebrate larvae. NI = compound has not been identified.

rebekka were screened for antimicrobial activity against 5 pathogenic bacteria (Wang et al. 2011). All isolates showed high to moderate antimicrobial activities, with the genera *Penicillium* and *Cladosporium* being the most active (Table 3). In another study, 121 fungal isolates were obtained from 6 species of healthy gorgonian corals from the South China Sea (Zhang et al. 2012b), and ~38% of the isolates inhibited the growth of environmental bacteria and pathogenic fungi. These facts suggest that epibiotic fungi potentially help their host to prevent microbial fouling.

Several antimicrobial compounds have been isolated from marine fungi (Table 3). The fungus *Ampelomyces* sp. produced 3-chloro-2,5-dihydroxybenzyl alcohol, which inhibited not only the growth of marine bacterial species, but also prevented settlement of *Hydroides elegans* and larvae of *Balanus amphitrite* (Kwong et al. 2006). The fungus *Aspergillus* sp. isolated from the sponge *Xestospongia testudinaria* produced 4 bisabolane-type sesquiterpenoids that were effective against environmental bacteria (Li et al. 2012). Similar to bacteria, the cultivation conditions of fungi seem to be an important factor in the production of bioactive compounds (Xiong et al. 2009).

Microbial disruption of biofilms

Microbial biofilms form in a multi-step process of signalling and regulatory events, all of which lead to the development of sessile microbial communities (in a physiological state that is distinct from planktonic cultures) encased in an extracellular polymer matrix. Therefore, microbial biofilms can be disrupted via interference with the signal exchange that leads to biofilm formation or through degradation of the various extracellular polymers that make up the matrices of the biofilm. In this review, the focus is on the *N*-acyl homoserine lactone (AHL)-based signalling system, as it is among the most intensively studied.

Enzymes that degrade biofilm-forming QS signals

Under laboratory conditions and in natural environments, the formation of a biofilm has been inhibited by enzymatic degradation of the signals involved in biofilm formation (Rajamani et al. 2011; Oh et al. 2012). Turn-over of bacterial AHLs is one of the best studied examples of the enzyme-mediated manipulation of bacterial signalling and biofilm formation. AHL-degrading microorganisms are taxonomically diverse and include true fungi (Ascomycetes and Basidiomycetes), Firmicutes and Gram-negative bacteria (Teplitski et al. 2011). Some bacteria can completely degrade AHLs through the combined action of several enzymes (Huang et al. 2003; Uroz et al. 2003, 2005). AHL-degrading enzymes include lactone hydrolases (lactonases), acylases and

oxidoreductases. The most studied AHL-lactonases belong to 2 families: Zn-dependent metallo- β -lactamases (eg the autoinducer inactivating enzyme (AiiA) of *Bacillus* sp. as a prototypical member (Dong et al. 2000)) and metallo-dependent phosphotriesterases (Uroz et al. 2008). Since the isolation of the first AHL-degrading enzyme, AiiA and the cloning of the gene *aiiA* encoding it from *Bacillus* sp. 240B1 (Dong et al. 2001), over 220 AiiA sequences from various *Bacillus* spp. have been deposited into the NCBI database.

A novel AHL-acylase was identified and isolated from the marine nitrogen-fixing filamentous cyanobacterium *Anabaena* sp. PCC 7120 (Romero et al. 2008). This enzyme, named 'autoinducer inhibitor from cyanobacteria' (AiiC), showed broad acyl chain length specificity and was produced during the exponential growth phase, but disappeared after 24 h. The AiiC enzyme could be used by the cyanobacterium to control the response of its own QS signals, but can potentially be used to interfere with signalling within mixed microbial communities. The search for the acylase sequences revealed the presence of similar genes in other cyanobacteria, such as *Nostoc punctiforme*, *N. violaceus* and *Synechocystis* sp., which indicate that these enzymes may be widespread among cyanobacteria (Romero et al. 2008).

Soon after AiiA was first identified, this enzyme or its various forms were expressed in other bacteria, plants and unicellular algae to test its applications for controlling virulence, disrupting biofilm formation and manipulating host-associated microbiota (Dong et al. 2001; Rajamani et al. 2011; Oh et al. 2012). In the unicellular alga *Chlamydomonas reinhardtii*, expression of the codon-optimised AiiA altered the composition of the alga-associated microbiota (Rajamani et al. 2011). When AiiA was expressed in *V. cholera*, it completely eradicated biofilm formation (Augustine et al. 2010). Transgenic *aiiA*-expressing bacteria have also been used to disrupt industrial biofilms: the introduction of micro-encapsulated transgenic *E. coli* strongly reduced biofouling of membranes in wastewater treatment facilities over an extended period of time (Oh et al. 2012). Several naturally occurring bacteria producing AHL-degrading enzymes have been identified as promising agents in reducing biofouling of membranes in wastewater treatment facilities (Oh et al. 2012).

Aside from their function in disrupting biofouling, AHL-degrading bacteria, isolated from healthy fish and aquatic invertebrates, have been shown to be promising biocontrol agents capable of disrupting the virulence of pathogenic *Vibrio* spp. (reviewed by Tinh et al. 2008; Cao et al. 2012). Thus, the identification of biological functions in marine microorganisms could have wide-reaching applications.

Disruption of biofilms with polymer-degrading enzymes

The extracellular matrix encasing monospecific and multispecies biofilms includes complex mixtures of polysaccharides, proteins and nucleic acids (Kaplan 2009). Marine microbes have been shown to disrupt biofilms through the production of lytic enzymes capable of degrading components of the biofilm matrix. For example, α -amylase from a marine isolate of *B. subtilis* reduced *Vibrio cholera* biofilms by 20–80% (Kalpana et al. 2012). Treatment of biofilms formed by human pathogens, eg *P. aeruginosa*, with α -amylase did not effectively reduce the ability of the pathogen to form a biofilm (Kalpana et al. 2012) as the major components of the biofilm matrix of pathogens (including *Vibrio* spp.) consist of glucose, galactose, *N*-acetylglucosamine and mannose (Fong et al. 2010). Treatment of biofilms containing poly- β -1,6-*N*-acetyl-D-glucosamine with Dispersin B (a β -*N*-acetyl-hexosaminidase) was effective in detaching pre-formed biofilms (Kaplan 2009). However, it is not known how common Dispersin B is in marine bacteria. A NCBI BLAST search revealed only one DspB homologue in the marine strain *Photobacterium* sp. SKA34 (Accession Number: SKA34_07164). On the other hand, the ability to degrade acetylated hexose amines and polymers containing them is widespread in marine bacteria (Riemann & Azam 2002; Krediet et al. 2009), but it is not yet known whether these *N*-acetyl-glycosaminidases have a role to play in biofilm dispersal. Furthermore, nucleic acids are becoming recognised as an important component of biofilms, including those formed by *V. cholerae* (Nijland et al. 2010; Seper et al. 2011). Self-produced and exogenous nucleases reduced biofilm formation by at least twofold (Kaplan 2009; Nijland et al. 2010; Seper et al. 2011). Therefore, a better understanding of the enzymatic activities by which marine bacteria dislodge existing biofilms will likely have further applications in the medical field and probably also for industry (Kaplan 2009).

Disruption of biofilms through inhibition of bacterial signalling and regulatory cascades

In addition to QS signals, such as AHLs, many other small molecules are known to trigger regulatory cascades leading to biofilm formation or dispersal (reviewed by Irie & Parsek 2008; Dobretsov et al. 2009). Inhibition or interference with QS regulation has been explored since the mid-1990s when the first mimics of bacterial AHLs were characterised in the red alga *D. pulchra*, which is itself mostly free of bacterial biofilms (Manefield et al. 1999). Screens of natural and synthetic products for QS inhibitory activities typically relied on biosensors, which are semi-synthetic constructs in which a reporter, typically genes involved in pigment production, fluorescence or luminescence, are inserted downstream from the

promoter responsive to AHLs. Such screens led to the identification of dozens of compounds with presumed QS-inhibitory activities. However, further work with 1 of the reporters (based on the *lasRI* QS system of *P. aeruginosa*) revealed that a number of small molecules produced by marine organisms can interfere with the activity of the reporter, although not only through interference with the receptor–AHL interactions (Dobretsov et al. 2010; Kwan et al. 2011).

In addition to compounds capable of modulating QS, several other small molecules with biofilm-inhibitory properties have been purified from marine bacteria. For example, 4-phenylbutanoic acid was isolated from a marine *Bacillus* sp., which was capable of strongly inhibiting the formation of biofilm by marine bacteria as well as by human pathogens (Nithya et al. 2011). Although the exact mechanism responsible for inhibiting the formation of biofilm remains unknown, the activity of the compound appears not be linked to an inhibition in the production of EPS (Nithya et al. 2011). Using *Vibrio vulnificus* EPS-defective mutants, it was found that D-glucosamine is responsible for inhibiting the formation of biofilm. It was also shown that glucosamine is inhibitory to the type-2 autoinducer system, yet glucosamine retained its inhibitive properties even after immobilisation on polymeric nanofibers (Kim et al. 2011).

Recently, indole has been identified as a signal molecule involved in the formation of biofilm by *V. cholerae* (Mueller et al. 2009), controlling the *vps* genes which regulate the production of *Vibrio* polysaccharides that make up the bulk of a biofilm matrix. Several natural products (containing indole moieties) extracted from a Mediterranean gorgonian coral were shown to be effective at inhibiting the formation of biofilms by 3 marine bacteria (Penez et al. 2011). Indole-producing bacteria also can inhibit the growth of pathogens (see antibacterial activity). As the genes controlled by indole are now characterised in *V. cholerae* (Mueller et al. 2009), it will be of interest to test whether the alkaloids produced by marine organisms are capable of interfering with bacterial indole-mediated signalling.

Cyanobacteria produce siderophores to enable them to obtain iron from the environment. Several types of cyanobacterial siderophores are known, such as schizokinen, synechobactins and anachelins (Goldman et al. 1983; Ito et al. 2004; Ito & Butler 2005). Recently, the catecholate fragment of the siderophore anachelin was obtained from the cyanobacterium *Anabaena cylindrica* and was used to immobilise polyethylene glycol (PEG) on surfaces in order to prevent the formation of biofilms (Gademann et al. 2009). This fragment was further modified by merging the anachelin chromophore with the antibiotic vancomycin through a PEG linker. This mixture was used for the generation of antimicrobial

surfaces on implants, catheters and stents, and thus has important medical applications.

Antialgal activity

Heterotrophic bacteria

In this review, antialgal activity is defined as the prevention of settlement of algal spores and diatom cells as well as inhibition of growth of diatoms and microalgae. Bacteria associated with the green alga *Ulva lactuca* have been shown to have antibacterial and antialgal activity (Table 4; Rao et al. 2007; Kumar et al. 2011). Eighty percent of epibiotic isolates from *U. lactuca* inhibited growth of the diatom *Cylindrotheca fusiformis* (Kumar et al. 2011). All isolates had differential activity; *Pseudoalteromonas* isolates were active against bacteria and diatoms, while members of the genera *Bacillus*, *Vibrio* and *Shewanella* demonstrated antidiatom activity only.

Bacterial isolates from seawater have been shown to have antialgal activities (Table 4). Biofilms of the bacterium *Alteromonas* sp. Ni1-LEM inhibited settlement of nine benthic diatoms and spores of *Ulva* sp. more efficiently than the reference strains *H. marina* and *Pseudolalteromonas tunicata* (Silva-Aciaries & Riquelme 2008). It was shown that the AF compound is thermostable and a hydrophilic protein >3500 Da is produced extracellularly by the bacterium. The bacterium *Pseudoalteromonas haloplanktis* CI4 was shown to produce a large (3–10 kDa), heat sensitive, non-protein compound that inhibited germination of zoospores of *U. pertusa* (Ma et al. 2010). Two AF steroids were isolated from the filamentous bacterium *Leucithrix mucor* (Cho 2012). These compounds inhibited settlement of zoospores of *U. pertusa* and the diatom *Navicula annexa*. The marine bacterium *Shewanella oneidensis* SCH0402 produced 2-hydroxymyristic acid and cis-9-oleic acid, which inhibited germination of spores of *U. pertusa* (Bhattarai et al. 2007). In field bioassays, these compounds were as efficient as the toxic AF compound tributyltin oxide.

Antilarval activity

Heterotrophic bacteria

Biofilms of many epibiotic bacteria inhibit settlement of invertebrate larvae in the laboratory (Table 5), but it is uncertain if the same bacteria exhibit this activity under natural conditions. Recent investigations provided evidence that some epibiotic biofilms could indeed prevent larval settlement on the surface of algae in natural conditions. For instance, *Pseudoalteromonas tunicata* and *Phaeobacter* sp. strain 2.10 (formerly *Roseobacter gallaeciensis*) isolated from the alga *U. australis* inhibited larval settlement of the bryozoan *Bugula neritina* at densities of 10^3 to 10^5 cells cm⁻², which are similar to

the densities of these bacteria under natural conditions (Rao et al. 2007). Multispecies biofilms with similar composition to those from the alga *Fucus vesiculosus* inhibited the settlement of cyprid larvae of the barnacle *Amphibalanus improvisus* (Nasrolahi et al. 2012).

Microbial biofilms affect adhesion of larvae and spores of fouling organisms. In the study of Zardus et al. (2008) settlement of the ascidian *Phallusia nigra*, the polychaete *H. elegans*, the bryozoan *B. neritina* and the barnacle *Balanus amphitrite* was investigated in the laboratory. Only adhesion of *B. neritina* larvae was unaffected, while adhesion of the other species was facilitated. In contrast, the bacterium *Cobetia marina* inhibited the adhesion strength of spores of *Ulva linza* (Mieszkina et al. 2012). It is not clear how bacteria facilitate or inhibit adhesion of other species but it is possible that multiple biochemical, behavioural and physical mechanisms may be involved (Zardus et al. 2008).

The AF activity of certain bacteria has been found to correlate with bacterial colour. This unexpected fact was described initially by Egan et al. (2002) for the bacterium *P. tunicata*. Purple mutants of *P. tunicata* lost AF activity because of a disruption of the gene *wmpR*, which is involved in the synthesis of AF compounds. Recently, a correlation between bacterial colour and AF activity has been shown for another bacterium, *Pseudoalteromonas* sp. sf57, which inhibits settlement of *H. elegans* larvae (Huang et al. 2011). Several mutants of this bacterium were made by transposon mutagenesis. White mutants formed thicker biofilms and became inductive to settlement by larvae. Further analysis suggested that the type II secretion pathway, the LysR transcriptional regulator, NAD(P)-binding proteins, flagella and cell membrane processes could play a role in the regulation of bioactivity in *Pseudoalteromonas* sp. sf57.

The ability of bacteria incorporated in a non-toxic matrix to inhibit larval settlement has been shown previously (Holmström et al. 2000). This 'living paint' contained the bacterium *P. tunicata* and was able to inhibit barnacle fouling for 14 days. A recent study with the same bacterial species demonstrated its ability to remain viable for up to 12 months when incorporated into a kappa-carrageenan matrix (Yee et al. 2007). Field experiments in Sydney harbour demonstrated the ability of these 'living paints', with *P. tunicata* able to inhibit fouling for up to 7 weeks. This suggests that by using the right matrix, it is possible to create a 'living paint' with a longer period of AF defence.

During the last 6 years, several AF compounds have been isolated and identified from marine bacteria (Table 5). These include 6 poly-ethers A–E isolated from the epibiotic bacterium *Winogradskyella poriferorum* (Dash et al. 2009, 2011). These natural products inhibited the settlement of the barnacle *B. amphitrite*, the polychaete *H. elegans* and the bryozoan *B. neritina*. The

Table 4. Inhibition of the settlement of algal spores and diatoms by marine heterotrophic bacteria.

Bacteria	Effective against	Source (host)	Location	Inhibitive compound	References	Comments
10 Bacterial isolates	<i>Cylindrotheca fusiformis</i>	<i>Ulva lactuca</i>	South Pacific Ocean, Fiji	NI	Kumar et al. (2011)	Species specific anti-diatom and anti-bacterial activity
<i>Alteromonas</i> sp.	Eight benthic diatoms and spores of <i>Ulva</i> sp.	Water column	South Pacific, Chile	Thermostable hydrophilic peptide	Silva-Aciares et al. (2008)	
<i>Leucothrix mucor</i>	<i>Navicula annexa</i> and spores of <i>Ulva pertusa</i>	Seawater	South Sea, Korea	17-(1,2-Dihydroxyl-5-methyl-hexane)-2,3-dihydroxyl-cholest-4-en-6-one [compound and 13-acetate-17-(1,5-dimethylhexane)-cholest-7-en-3,5,6,15-tetraol	Cho (2012)	
<i>Shewanella oneidensis</i> SCH0402	Spores of <i>Ulva pertusa</i> ; barnacles; mussels	Seawater	Yellow Sea, Korea	2-Hydroxymyristic acid and cis-9-oleic acid	Bhattarai et al. (2007)	Field experiments showed good AF effect after 1.5 years
22 Bacterial isolates	Spores of <i>Ulva</i> sp.	Water column	Baltic Sea	NI	Bembom et al. (2011)	Strongest activity among <i>Pseudoalteromonas</i> strains
<i>Pseudoalteromonas haloplanktis</i> C14	Spores of <i>U. pertusa</i>	Seawater	South China Sea	Heat sensitive 3–10 kDa, not protein	Ma et al. (2010)	
<i>Pseudoalteromonas tunicata</i> and <i>Phaeobacter</i> sp. strain 2.10	<i>Polysiphonia</i> sp.	<i>Ulva australis</i>	Pacific Ocean	NI	Rao et al. (2007)	

Note: Bacterial strains are classified by their activity. NI = compound has not been identified.

Table 5. Inhibition of the settlement of invertebrate larvae by marine bacteria.

Bacteria	Effective against	Source (host)	Location	Inhibitive compound	Reference	Comments
Heterotrophic bacteria						
<i>Pseudoalteromonas tunicata</i> and <i>Phaeobacter</i> sp. strain 2.10	<i>Bugula neritina</i>	<i>Ulva australis</i>	Pacific Ocean	NI	Rao et al. (2007)	
<i>Pseudoalteromonas issachenkonii</i> UST041101–043	<i>B. neritina</i>	Deep sea sediments	Pacific Ocean	Protease	Dobretsov et al. (2007)	Active in laboratory and field experiments
Natural communities, <i>Photobacterium halotolerans</i> and <i>Ulvibacter litoralis</i> , <i>Shewanella basaltis</i>	<i>Amphibalanus improvisus</i>	<i>Fucus cetratus</i> , <i>F. vesiculosus</i> and <i>Polysiphonia stricta</i>	Baltic Sea	NI	Nasrolahi et al. (2012)	
<i>Pseudoalteromonas arctica</i> , <i>Shewanella baltica</i> and <i>Bacillus foraminis</i>						
<i>Streptomyces fungicidicus</i>	<i>Balanus amphitrite</i>	Deep sea sediments	West Pacific Ocean	Five Diketopiperazines	Li et al. (2006)	
<i>Pseudomonas rhizosphaerae</i>	<i>B. amphitrite</i> ; <i>B. neritina</i>	Deep sea	Pacific Ocean Province A	Bis(2-ethylhexyl)phthalate, cyclo (Tyr-Ile), cyclo(Phe-Pro), cyclo (Val-Pro) 3-phenyl-2-propenoic acid	Qi, Xu, Gao (2009)	Both antimicrobial and antilarval activities
<i>Pseudoalteromonas</i> sp. sf57	<i>Hydroides elegans</i>	Natural biofilms	South China Sea	NI	Huang et al. (2011)	Inhibition correlates with bacterial pigmentation
<i>Streptomyces</i> sp. UST040711–290	<i>H. elegans</i>	Deep sea sediments	Pacific Ocean Province A	Fatty acid 12-methyltetradecanoid acid	Xu et al. (2009)	Demonstrated effect of this compound on larval gene expression
<i>Winogradskyella poriferorum</i>	<i>H. elegans</i> ; <i>B. amphitrite</i>	<i>Lissodendoryx isodictyalis</i>	Caribbean, Atlantic Ocean	Poly-ether A-E	Dash et al. (2009); Dash et al. (2011)	Inhibitive concentrations of 70 and 60 $\mu\text{g ml}^{-1}$
<i>Streptomyces</i> sp.	<i>H. elegans</i> ; <i>B. neritina</i> and <i>B. amphitrite</i>	Deep sea sediments, seawater	West Pacific Ocean	Five compounds belonging to α , β unsaturated lactones	Xu et al. (2010)	Based on structure relations antifouling compound was synthesised
<i>Macrococcus</i> sp., <i>Bacillus</i> sp., <i>Pseudoalteromonas</i> sp.	<i>Aulacomya maoritana</i>	Biofilms	South-West Pacific Ocean, New Zealand	NI	Alfaro et al. (2011)	Highest inhibition for bacterial exudates

thiopalmirone and palmyrrolinone, and they were shown to inhibit the growth of the snail *B. glabrata* ($LC_{50}=8.3$ and $6.0 \mu\text{M}$, respectively). Another molluscicidal agent against *B. glabrata* identified as Cyanolide A was isolated from *Lyngbya bouillondii*, a cyanobacterium originating from Papua New Guinea (Pereira et al. 2010).

L. majuscula is a marine cyanobacterium with a wide distribution. The species has recently been studied for the production of several bioactive compounds including those with AF activity (Paul et al. 2011). More than 113 bioactive compounds were isolated from this organism, out of which 10% possessed antifungal and antimicrobial activity (Burja et al. 2001). Extracts of *L. majuscula* isolated from the western lagoon of Pulau Hantu, Singapore were incorporated into a PhytigelTM matrix and the development of fouling communities was followed over 4 weeks (Tan & Goh 2009). The data showed that extracts from *L. majuscula* significantly inhibited the settlement of barnacles. Based on a settlement assay using the barnacle *Amphibalanus (Balanus) amphitrite*, a number of AF products were purified and identified as isolmajusculamide, majusculamide A and hanutupeptin C (Tan & Goh 2009).

Fungi

Several AF compounds have been isolated recently from marine fungi (Table 3). Two fungal species, *Cochliobolus lunatus* (Shao, Wang et al. 2011) and *Aspergillus* sp. (Shao, Wu et al. 2011), were isolated from the gorgonian coral *Dichotella gemmacea*. The fungus *C. lunatus* was shown to produce 14 new resorcylic acid lactones that inhibited the settlement of cyprid larvae of the barnacle *B. amphitrite*. The fungus *Aspergillus* sp. produces aspergilones A that inhibited the settlement of *B. amphitrite* with an EC_{50} value of $7.68 \mu\text{g ml}^{-1}$ (Shao, Wu et al. 2011). Penispirolloid A produced by *Penicillium* sp. OUCMDZ-776 isolated from hypersaline sediment in China inhibited the settlement of larvae of *B. neritina* with an EC_{50} of $2.40 \mu\text{g ml}^{-1}$ (He et al. 2012). The marine fungus *Cladosporium* sp. F14 produced 3-phenyl-2propenoic acid and bis(2-ethylhexyl)phthalate, which effectively inhibited the settlement of *B. neritina* and *B. amphitrite* (Qi, Xu, Xiong et al. 2009).

Looking ahead

Epibiotic microorganisms associated with marine algae, sponges, corals and other invertebrates have high AF potential. Whilst the question of whether these epibiotic microorganisms protect their hosts from biofouling remains open, without doubt, epibiotic microorganisms remain a potential target for the isolation of novel AF compounds. Since the review of Dobretsov et al. (2006), a number of AF compounds have been isolated from deep sea bacteria and marine fungi. These groups of

microorganisms deserve further investigation, especially considering the fact that several patents based on their AF properties have been filed (Melde & Markowitz 2012; Qian et al. 2012).

It is clear that some compounds produced by microorganisms can have multiple functions. For example, indole is involved in the formation of biofilms by *V. cholerae* (Mueller et al. 2009) and has significant activity against bacterial pathogens (Devi et al. 2010). DKPs produced by many bacteria can work as signal molecules (Tommonaro et al. 2012), inhibit the formation of biofilms (de Carvalho & Abraham 2012), the growth of bacteria (Qi, Xu, Xiong et al. 2009) and the settlement of larvae (Li et al. 2006; Qi, Xu, Gao et al. 2009). Compounds which inhibit both microfouling and macrofouling should be investigated further as future potential AF agents.

Isolation of compounds from multispecies communities, such as microbial mats and complex biofilms, not from single isolates, is a potential new way of obtaining novel AF compounds. This approach is based on the fact that some bioactive compounds are only produced by microorganisms when in mixed cultures or consortia. The challenges of this method include difficulties in obtaining AF compounds in sufficiently large quantities and changes in the production of compounds due to changes in the microbial communities. Additionally, antifoulants might not be effective against microbes present in the same biofilm. In spite of these drawbacks, it was shown that cyanobacterial mats from hot springs and biological desert crusts produce antibacterial and QS inhibitory compounds under *in situ* conditions (Abed et al. 2011; Dobretsov, Abed et al. 2011). It is of note that bioactivity in the microbial mats was correlated with the diversity of the microbial communities; extracts of less diverse microbial mats from hot springs had higher antibacterial activity than the extracts of more diverse mats (Dobretsov, Abed et al. 2011).

The AF properties of dozens of marine compounds are currently being characterised and it is apparent that many more are still to be discovered. Even though many compounds can inhibit biofouling in laboratory conditions, very little is known about their performance in the field (Dobretsov, Teplitski et al. 2011). From an ecological point of view, it is important to learn whether these compounds are produced *in situ* and whether they are produced in quantities that are sufficient to trigger meaningful responses in marine organisms. The question regarding the production of bioactive compounds *in situ* by bacteria is especially relevant, since, unlike macroorganisms and filamentous bacteria, collecting 'wild' samples of bacteria in the volumes suitable for chemical isolation and identification is not a trivial undertaking. It is also commonly accepted that the production of bioactive compounds by organisms is greatly affected by the

presence of specific neighbouring species, nutrients, signals and environmental factors (eg temperature, salinity and light). Therefore, establishing whether AF compounds are produced *in situ* seems especially important for understanding their functions. The deployment of 'chemical traps' (essentially dialysis pouches filled with hydrophobic resins) has helped to demonstrate that compounds inhibitory to biofilms are present within the mucus-associated microbial communities on the surface of corals (Alagely et al. 2011), and although the chemical nature of these activities was not determined, this was the first report of the presence of such bioactivity on the surface of healthy corals.

Recent advances in molecular tools will facilitate the analysis of the diversity and function of fouling microbial communities, which will help in designing strategies to reduce their development. Pyrosequencing has the advantage of creating thousands of sequences read at a very low cost and thus provides a comprehensive overview of the structure of fouling communities. When this technique is combined with other metabolomic, transcriptomic and proteomic techniques, more information will be obtained on the metabolism and interaction among different groups of organisms within fouling communities. The identification of transcripts (transcriptomic) and polypeptides (proteomics) differentially accumulated in response to a particular AF compound will help restrict the potential molecular targets of the chemicals of interest. Once it is established that the activities of interest are produced *in situ* and have an effect at meaningful concentrations, it should be feasible to test how the composition of microbial communities changes in the presence of the AF activities. Other techniques, such as Nano-SIMS and Raman spectroscopy, can be used to study the interaction between organisms in field samples as well as in model cultures. Recent developments in single-cell technologies including sorting individual cells and sequencing their genomes should be useful in exploring the genetic makeup of chemically rich microorganisms. Relevant genes that code for certain AF compounds may be cloned and optimised for commercial production.

Several studies have demonstrated the usefulness of natural compounds and enzymes in preventing and disrupting biofilms and dozens of patents have been filed. However, to date these compounds are not widely used in controlling biofouling on industrial scales. Clearly, extraction of marine micro- and macro-organisms cannot provide a sustainable and cost-effective method of obtaining pure compounds suitable for industry. Therefore, the production of natural products derived from microbes has to be through scale-up in bioreactors or by chemical synthesis. Questions regarding the toxicity of compounds to other marine organisms also need to be addressed prior to wider commercial application. Overall, this review shows that marine microorganisms are an

important source of antibacterial, antialgal and antilarval compounds, which can potentially be used for solving the biofouling problems of industry and for other biotechnological applications.

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References

- Abdelmohsen UR, Pimentel-Elardo SM, Hanora A, Radwan M, Abou-El-Ela SH, Ahmed S, Hentschel U. 2010. Isolation, phylogenetic analysis and anti-infective activity screening of marine sponge-associated actinomycetes. *Mar Drug*. 8:399–412.
- Abed RMM, Dobretsov S, AL-Kharusi S, Jupp B, Golubic S. 2011. Microscopic diversity and bioactivity of an inland extremely hypersaline cyanobacterial mats from a desert stream in the Sultanate of Oman. *Fottea*. 11:215–224.
- Abed RMM, Dobretsov S, Sudesh K. 2009. Applications of cyanobacteria in biotechnology. *J Appl Microbiol*. 106:1–12.
- Alagely A, Krediet CJ, Ritchie KB, Teplitski M. 2011. Signaling-mediated cross-talk modulates swarming and biofilm formation in a coral pathogen *Serratia marcescens*. *ISME J*. 5:1609–1620.
- Alfaro AC, Young T, Ganesan AM. 2011. Regulatory effects of mussel (*Aulacomya maoriana* Iredale 1915) larval settlement by neuroactive compounds, amino acids and bacterial biofilms. *Aquaculture*. 322–323:158–168.
- Augustine N, Kumar P, Thomas S. 2010. Inhibition of *Vibrio cholerae* biofilm by AiiA enzyme produced from *Bacillus* spp. *Arch Microbiol*. 192:1019–1022.
- Bernbom N, Ng YY, Kjelleberg S, Harder T, Gram L. 2011. Marine bacteria from Danish coastal waters show antifouling activity against the marine fouling bacterium *Pseudomonas* sp. strain S91 and zoospores of the green alga *Ulva australis* independent of bacteriocidal activity. *Appl Environ Microbiol*. 77:8557–8567.
- Bhattacharai HD, Ganti VS, Paudel B, Lee YK, Lee HK, Hong YK, Shin HW. 2007. Isolation of antifouling compounds from the marine bacterium, *Shewanella oneidensis* SCH0402. *World J Microb Biotech*. 23:243–249.
- Biondi N, Tredici MR, Taton A, Wilmotte A, Hodgson DA, Losi D, Marinelli F. 2008. Cyanobacteria from benthic mats of Antarctic lakes as a source of new bioactivities. *J Appl Phycol*. 105:105–115.
- Briand JF, Djeridi I, Jamet D, Coupé S, Bressy C, Molmeret M, Le Berre B, Rimet F, Bouchez A, Blache Y. 2012. Pioneer marine biofilms on artificial surfaces including antifouling coatings immersed in two contrasting French Mediterranean coast sites. *Biofouling*. 28:453–463.
- Brown GD, Wong H-F, Hutchinson N, Lee S-C, Chan BKK, Williams GA. 2004. Chemistry and biology of maculalactone A from the marine cyanobacterium *Kyrtuthrix maculans*. *Phytochem Rev*. 3:381–400.

- Bruhn JB, Gram L, Belas R. 2007. Production of antibacterial compounds and biofilm formation by *Roseobacter* species are influenced by culture conditions. *Appl Environ Microbiol.* 73:442–450.
- Burja AM, Banaigs B, Abou-Mansour E, Burgess JG, Wright PC. 2001. Marine cyanobacteria – a prolific source of natural products. *Tetrahedron.* 57:9347–9377.
- Caicedo NH, Heyduck-Soller B, Fischer U, Thoming J. 2011. Bioproduction of antimicrobial compounds by using marine filamentous cyanobacterium cultivation. *J Appl Phycol.* 23:811–818.
- Cao YA, He SX, Zhou ZG, Zhang MC, Mao W, Zhang HT, Yao B. 2012. Orally administered thermostable N-Acyl homoserine lactonase from *Bacillus* sp. strain AI96 attenuates *Aeromonas hydrophila* infection in zebrafish. *Appl Environ Microbiol.* 78:1899–1908.
- Cassé F, Swain GW. 2006. The development of microfouling on four commercial antifouling coatings under static and dynamic immersion. *Int Biodeterior Biodegr.* 57:179–185.
- Chen YH, Kuo J, Sung PJ, Chang YC, Lu MC, Wong TY, Liu JK, Weng CF, Twan WH, Kuo FW. 2012. Isolation of marine bacteria with antimicrobial activities from cultured and field-collected soft corals. *World J Microbiol Biotech.* 28:3269–3279.
- Cho JY. 2012. Antifouling steroids isolated from red alga epiphyte filamentous bacterium *Leucothrix mucor*. *Fish Sci.* 78:683–689.
- Dahms H-U, Xu Y, Pfeiffer C. 2006. Antifouling potential of cyanobacteria: a mini-review. *Biofouling.* 22:317–327.
- Dash S, Jin C, Lee OO, Xu Y, Qian PY. 2009. Antibacterial and antilarval-settlement potential and metabolite profiles of novel sponge-associated marine bacteria. *J Ind Microb Biotechnol.* 36:1047–1056.
- Dash S, Nogata Y, Zhou X, Zhang Y, Xu Y, Guo X, Zhang X, Qian PY. 2011. Poly-ethers from *Winogradskyella poriferorum*: antifouling potential, time-course study of production and natural abundance. *Bioresour Technol.* 102:7532–7537.
- De Carvalho MP, Abraham WR. 2012. Antimicrobial and biofilm inhibiting diketopiperazines. *Curr Med Chem.* 19:3564–3577.
- Devi P, Wahidullah S, Rodrigues C, Souza LD. 2010. The sponge-associated bacterium *Bacillus licheniformis* SAB1: a source of antimicrobial compounds. *Mar Drugs.* 8:1203–1212.
- Dheilly A, Soum-Soutera E, Klein GL, Bazire A, Compere C, Haras D, Dufour A. 2010. Antibiofilm activity of the marine bacterium *Pseudolateromonas* strains 3J6. *Appl Environ Microbiol.* 76:3452–3461.
- Dobretsov S. 2010. Marine Biofilms. In: Dürr S, Thomason J, editors. *Biofouling*. Oxford: Wiley-Blackwell; p. 123–136.
- Dobretsov S, Abed RMM, Al Maskari SMS, Al Sabahi JN, Victor R. 2011. Cyanobacterial mats from hot springs produce antimicrobial compounds and quorum sensing inhibitors under natural conditions. *J Appl Phycol.* 23:983–993.
- Dobretsov S, Dahms HU, Qian PY. 2006. Inhibition of biofouling by marine microorganisms and their metabolites. *Biofouling.* 22:43–54.
- Dobretsov S, Teplitski M, Alagely A, Gunasekera SP, Paul VJ. 2010. Malyngolide from the cyanobacterium *Lyngbya majuscula* interferes with quorum sensing circuitry. *Environ Microbiol Rep.* 2:739–744.
- Dobretsov S, Teplitski M, Bayer M, Gunasekera S, Proksch P, Paul VJ. 2011. Inhibition of marine biofouling by bacterial quorum sensing inhibitors. *Biofouling.* 27:893–905.
- Dobretsov S, Teplitski M, Paul V. 2009. Mini-review: quorum sensing in the marine environment and its relationship to biofouling. *Biofouling.* 25:413–427.
- Dobretsov S, Thomason J. 2011. The development of marine biofilms on two commercial non-biocidal coatings: a comparison between silicone and fluoropolymer technologies. *Biofouling.* 27:869–880.
- Dobretsov S, Xiong H, Xu Y, Levin LA, Qian PY. 2007. Novel antifoulants: inhibition of larval attachment by proteases. *Mar Biotech.* 9:388–397.
- Dong YH, Wang LH, Xu JL, Zhang HB, Zhang XF, Zhang LH. 2001. Quenching quorum-sensing-dependent bacterial infection by an N-acyl homoserine lactonase. *Nature.* 411:813–817.
- Dong YH, Xu JL, Li XZ, Zhang LH. 2000. AiiA, an enzyme that inactivates the acylhomoserine lactone quorum-sensing signal and attenuates the virulence of *Erwinia carotovora*. *Proc Natl Acad Sci USA.* 97:3526–3531.
- Egan S, Hader T, Burke C, Steinberg P, Kjelleberg S, Thomas T. 2012. The seaweed holobiont: understanding seaweed-bacteria interactions. *FEMS Microb Rev.* doi: 10.1111/1574-6976.12011. Available from: <http://onlinelibrary.wiley.com/doi/10.1111/1574-6976.12011/supinfo>
- Egan S, James S, Kjelleberg S. 2002. Identification and characterization of a putative transcriptional regulator controlling the expression of fouling inhibitors in *Pseudoalteromonas tunicata*. *Appl Environ Microbiol.* 68:372–378.
- Egan S, Thomas T, Kjelleberg S. 2008. Unlocking the diversity and biotechnological potential of marine surface associated microbial communities. *Curr Opin Microbiol.* 11:219–225.
- Fong JC, Syed KA, Klose KE, Yildiz FH. 2010. Role of *Vibrio polysaccharide* (vps) genes in VPS production, biofilm formation and *Vibrio cholerae* pathogenesis. *Microbiol.* 156:2757–2769.
- Fusetani N. 2011. Antifouling marine natural products. *Nat Prod Rep.* 28:400–410.
- Gademann K, Kobylinska J, Wash J-Y, Woods TM. 2009. Surface modifications based on the cyanobacterial siderophore anachelin: from structure to functional biomaterials design. *Biomaterials.* 22:595–604.
- Gantar M, Kaczmarek LT, Stanic D, Miller AW, Richardson LL. 2011. Antibacterial activity of marine and black band disease cyanobacteria against coral-associated bacteria. *Mar Drug.* 9:2089–2105.
- Golberg K, Eltzov E, Shnit-Orland M, Marks RS, Kush-Maró A. 2011. Characterization of quorum sensing signals in coral-associated bacteria. *Microb Ecol.* 61:783–792.
- Goldman SJ, Lammers PJ, Berman MS, Sanders-Loehr J. 1983. Siderophore-mediated iron uptake in different strains of *Anabaena* sp. *J Bacteriol.* 156:1144–1150.
- Gram L, Melchiorson J, Bruhn JB. 2010. Antibacterial activity of marine culturable bacteria collected from a global sampling of ocean surface waters and surface swabs of marine organisms. *Mar Biotech.* 12:439–451.
- Hadfield M. 2010. Biofilms and marine invertebrate larvae: what bacteria produce that larvae use to choose settlement sites. *Ann Rev Mar Sci.* 3:453–470.
- Harder T, Campbell AH, Egan S, Steinberg PD. 2012. Chemical mediation of ternary interactions between marine holobionts and their environment as exemplified by the red alga *Delisea pulchra*. *J Chem Ecol.* 38:442–450.
- Hayashiga-Soiza G, Uchida A, Mori N, Kuwahara Y, Ishida Y. 2008. Purification and characterization of antibacterial substances produced by a marine bacterium *Pseudoalteromonas haloplanktis* strain. *J Appl Microbiol.* 105:1672–1677.

- He F, Liu Z, Yang J, Fu P, Peng J, Zhu W-M, Qi SH. 2012. A novel antifouling alkaloid from halotolerant fungus *Penicillium* sp. OUCMDZ-776. *Tetrahedron Lett.* 53:2280–2283.
- Heindl H, Wiese J, Thiel V, Imhoff JF. 2010. Phylogenetic diversity and antimicrobial activities of bryozoan-associated bacteria isolated from Mediterranean and Baltic Sea habitats. *Syst Appl Microb.* 33:94–104.
- Holmström C, Steinberg P, Christov V, Christie G, Kjelleberg S. 2000. Bacteria immobilised in gels: improved methodologies for antifouling and biocontrol applications. *Biofouling.* 15:109–117.
- Huang JJ, Han JI, Zhang LH, Leadbetter JR. 2003. Utilization of acyl-homoserine lactone quorum signals for growth by a soil pseudomonad and *Pseudomonas aeruginosa* PAO1. *Appl Environ Microbiol.* 69:5941–5949.
- Huang YL, Ki J-S, Lee OO, Qian P-Y. 2009. Evidence for the dynamics of Acyl homoserine lactone and AHL-producing bacteria during subtidal biofilm formation. *ISME J.* 3:296–304.
- Huang YL, Li M, Yu ZL, Qian PY. 2011. Correlation between pigmentation and larval settlement deterrence by *Pseudoalteromonas* sp. sf57. *Biofouling.* 27:287–293.
- Irie Y, Parsek MR. 2008. Quorum sensing and microbial biofilms. *Curr Top Microbiol Immunol.* 322:67–84.
- Isnansetyo A, Kamei Y. 2009. Bioactive substances produced by marine isolates of *Pseudomonas*. *J Ind Microbiol Biotechnol.* 36:1239–1248.
- Ito Y, Butler A. 2005. Structure of synechobactins, new siderophores of the marine cyanobacterium *Synechococcus* sp. PCC 7002. *Limnol Oceanogr.* 50:1918–1923.
- Ito Y, Ishida K, Okada S, Murakami M. 2004. The absolute stereochemistry of anachelins, siderophores from the cyanobacterium *Anabaena cylindrica*. *Tetrahedron.* 60:9075–9080.
- Jaki B, Orjala J, Heilmann J, Linden A, Vogler B, Sticher O. 2000. Novel extracellular diterpenoids with biological activity from the cyanobacterium *Nostoc commune*. *J Nat Prod.* 63:339–343.
- James G, Holmström C, Kjelleberg S. 1996. Purification and characterization of a novel antibacterial protein from the marine bacterium D2. *Appl Environ Microbiol.* 62:2783–2783.
- Jebasingh SEJ, Murugan A. 2011. Antagonistic activity of the barnacle (*Balanus amphitrite*) associated bacteria against human bacterial pathogens. *World J Med Sci.* 6:36–41.
- Kalpna BJ, Aarthy S, Pandian SK. 2012. Antibiofilm activity of alpha-amylase from *Bacillus subtilis* S8–18 against biofilm forming human bacterial pathogens. *Appl Biochem Biotechnol.* 167:1778–1794.
- Kanagasabhapathy M, Nagata S. 2008. Cross-species induction of antibacterial activity produced by epibiotic bacteria isolated from Indian marine sponge *Pseudoceratina purpurea*. *World J Microb Biotechnol.* 24:687–691.
- Kaplan JB. 2009. Therapeutic potential of biofilm-dispersing enzymes. *Int J Artif Organs.* 32:545–554.
- Kaushik P, Garima CA, Goyal P. 2009. Screening of *Lyngbya majuscula* for potential antibacterial activity and HPTLC analysis of active methanolic extract. *J Pure Appl Microbiol.* 3:169–174.
- Kennedy J, Baker P, Piper C, Cotter PD, Walsh M, Mooij MJ, Bourke MB, Rea MC, O'Connor PM, Ross RP, et al. 2009. Isolation and analysis of bacteria with antimicrobial activities from the marine sponge *Haliclona simulans* collected from Irish waters. *Mar Biotechnol.* 11:384–396.
- Kim M, Park JM, Um HJ, Lee KH, Kim H, Min J, Kim YH. 2011. The antifouling potentiality of galactosamine characterized from *Vibrio vulnificus* exopolysaccharide. *Biofouling.* 27:851–857.
- Krediet CJ, Ritchie KB, Teplitski M. 2009. Catabolite regulation of enzymatic activities in a white pox pathogen and commensal bacteria during growth on mucus polymers from the coral *Acropora palmata*. *Dis Aquatic Org.* 87:57–66.
- Kumar V, Rao D, Thomas T, Kjelleberg S, Egan S. 2011. Antidiatom and antibacterial activity of epiphytic bacteria isolated from *Ulva lactuca* in tropical waters. *World J Microbiol Biotech.* 27:1543–1549.
- Kwan JC, Meickle T, Ladwa D, Teplitski M, Paul V, Luesch H. 2011. Lyngbyoic acid, a “tagged” fatty acid from a marine cyanobacterium, disrupts quorum sensing in *Pseudomonas aeruginosa*. *Mol Biosystems.* 7:1205–1216.
- Kwong TF, Miao L, Li X, Qian PY. 2006. Novel antifouling and antimicrobial compound from a marine-derived fungus *Ampelomyces* sp. *Mar Biotechnol.* 8:634–640.
- Landoulsi J, Cooksey KE, Dupres V. 2011. Review – interactions between diatoms and stainless steel: focus on biofouling and biocorrosion. *Biofouling.* 27:1105–1124.
- Lewandowski Z. 2000. Structure and function of biofilms. In: Evans LV, editor. *Biofilms: recent advances in their study and control*. Amsterdam: Harwood Academic; p. 1–17.
- Li D, Xu Y, Shao CL, Yang RY, Zheng CJ, Chen YY, Fu XM, Qian PY, She ZG, de Voogd NJ, Wang CY. 2012. Antibacterial bisabolane-type sesquiterpenoids from the sponge-derived fungus *Aspergillus* sp. *Mar Drugs.* 10:234–241.
- Li X, Dobretsov S, Xu Y, Xiao X, Hung OS, Qian P-Y. 2006. Antifouling diketopiperazines produced by a deep-sea bacterium, *Streptomyces fungicidicus*. *Biofouling.* 22:201–208.
- Ma YX, Liu PL, Zhang YS, Cao SM, Li DT, Chen W. 2010. Inhibition of spore germination of *Ulva pertusa* by the marine bacterium *Pseudolateromonas haloplanktis* CI4. *Acta Oceanol Sinica.* 29:69–78.
- Manefield M, de Nys R, Kumar N, Read R, Givskov M, Steinberg P, Kjelleberg S. 1999. Evidence that halogenated furanones from *Delisea pulchra* inhibit acylated homoserine lactone (AHL)-mediated gene expression by displacing the AHL signal from its receptor protein. *Microbiol.* 145:283–291.
- Mangano S, Michaud L, Caruso C, Brilli M, Bruni V, Fani R, Lo GA. 2009. Antagonistic interactions between psychrotrophic cultivable bacteria isolated from Antarctic sponges: a preliminary analysis. *Res Microbiol.* 160:27–37.
- Martins RF, Ramos MF, Herfindal L, Sousa JA, Skærven K, Vasconcelos VM. 2008. Antimicrobial and cytotoxic assessment of marine cyanobacteria – *Synechocystis* and *Synechococcus*. *Mar Drugs.* 6:1–11.
- Matz C, Webb JS, Schupp PJ, Phang SY, Penesyan A, Egan S, Steinberg P, Kjelleberg S. 2008. Marine biofilm bacteria evade eukaryotic predation by targeted chemical defense. *PLoS ONE.* 3:e2744.
- Melde B, Markowitz MA, inventors; Naval Research Laboratory, assignee. 2012. Composite for controlled release of small molecules in aquatic environments. United States patent US 20130011459.
- Mieszkina S, Tanchereau P, Callow ME, Callow JA. 2012. Effect of bacterial biofilms formed on fouling-release coatings from natural seawater and *Cobetia marina*, on the adhesion of two marine alga. *Biofouling.* 28:953–968.

- Molino PJ, Childs S, Eason Hubbard MR, Carey JM, Burgman MA, Wetherbee R. 2009. Development of the primary bacterial microfouling layer on antifouling and fouling release coatings in temperate and tropical environments in Eastern Australia. *Biofouling*. 25:149–162.
- Montaser R, Paul VJ, Luesch H. 2011. Pitiptolides C-F, antimycobacterial cyclodepsipeptides from the marine cyanobacterium *Lyngbya majuscula* from Guam. *Phytochemistry*. 72:2068–2074.
- Mueller RS, Beyhan S, Saini SG, Yildiz FH, Bartlett DH. 2009. Indole acts as an extracellular cue regulating gene expression in *Vibrio cholerae*. *J Bacteriol*. 191:3504–3516.
- Muscholl-Silberhorn A, Thiel V, Imhoff JF. 2008. Abundance and bioactivity of cultured sponge-associated bacteria from the Mediterranean Sea. *Microb Ecol*. 55:94–106.
- Nasrolahi A, Stratil SB, Jacob KJ, Wahl M. 2012. A protective coat of microbes on macroalgae: inhibitory effects of bacterial biofilms and epibiotic microbial assemblages on barnacle attachment. *FEMS Microbiol Ecol*. 81:583–595.
- Nijland R, Hall MJ, Burgess JG. 2010. Dispersal of biofilms by secreted, matrix degrading, bacterial DNase. *PLoS ONE*. 5:e15668.
- Nithya C, Devi MG, Pandian KS. 2011. A novel compound from the marine bacterium *Bacillus pumilus* S6–15 inhibits biofilm formation in Gram-positive and Gram-negative species. *Biofouling*. 27:519–528.
- Oh HS, Yeon KM, Yang CS, Kim SR, Lee CH, Park SY, Han JY, Lee JK. 2012. Control of membrane biofouling in MBR for wastewater treatment by quorum quenching bacteria encapsulated in microporous membrane. *Environ Sci Tech*. 46:4877–4884.
- Ortega-Morales BO, Chan-Bacab MJ, Mirnada-Tello E, Fardeau ML, Carrero JC, Stein T. 2008. Antifouling activity of sessile bacilli derived from marine surfaces. *J Industr Microb Biotech*. 35:9–15.
- Paul VJ, Ritson-Williams R. 2008. Marine chemical ecology. *Nat Prod Rep*. 25:662–695.
- Paul VJ, Ritson-Williams R, Sharp K. 2011. Marine chemical ecology in benthic environments. *Nat Prod Rep*. 28:345–387.
- Penesyan A, Kjelleberg S, Egan S. 2010. Development of novel drugs from marine surface associated microorganisms. *Mar Drugs*. 8:438–459.
- Penesyan A, Marshall-Jones Z, Holmström C, Kjelleberg S, Egan S. 2009. Antimicrobial activity observed among cultured epiphytic bacteria reflects their potential as source of new drugs. *FEMS Microbiol Ecol*. 69:113–124.
- Penesyan A, Tebben J, Lee M, Thomas T, Kjelleberg S, Harder T, Egan S. 2011. Identification of the antibacterial compound produced by the marine epiphytic bacterium *Pseudovibrio* sp. D323 and related sponge-associated bacteria. *Mar Drugs*. 9:1391–1402.
- Penez N, Culioli G, Perez T, Briand JF, Thomas OP, Blache Y. 2011. Antifouling properties of simple indole and purine alkaloids from the Mediterranean gorgonian *Paramuricea clavata*. *J Nat Prod*. 74:2304–2308.
- Pereira AR, Ertzbach L, Engene N, Muller R, Gerwick WH. 2011. Molluscicidal metabolites from an assemblage of Palmyra Atoll cyanobacteria. *J Nat Prod*. 74:1175–1181.
- Pereira AR, Mccue CF, Gerwick WH. 2010. Cyanolide A, a glycosidic macrolide with potent molluscicidal activity from the Papua New Guinea. *J Nat Prod*. 73:217–220.
- Pulz O, Gross W. 2004. Valuable products from biotechnology of microalgae. *Appl Microbiol Biotechnol*. 65:635–648.
- Qi SH, Xu Y, Gao J, Qian P-Y, Zhang S. 2009. Antibacterial and antilarval compounds from marine bacterium *Pseudomonas rhizosphaerae*. *Annal Microb*. 59:229–233.
- Qi SH, Xu Y, Xiong HR, Qian PY, Zhang S. 2009. Antifouling and antibacterial compounds from a marine fungus *Cladosporium* sp. F14. *World J Microb Biotech*. 25:399–406.
- Qian P-Y, Lau SCK, Dahms H-U, Dobretsov S, Harder T. 2007. Marine biofilm as mediator of colonization by marine macroorganisms: implications for antifouling and aquaculture. *Mar Biotech*. 9:399–410.
- Qian P-Y, Li X, Kwong FN, Yang LH, Dobretsov SV, inventors; St. Joseph' Children' Hospital, assignee. 2012 Feb 14. Use of marine fungus originated compounds as antifouling agents. United States patent US8114657.
- Qian P-Y, Xu Y, Fusetani N. 2010. Natural products as antifouling compounds: recent progress and future perspectives. *Biofouling*. 223–234.
- Rajamani S, Teplitski M, Kumar A, Krediet CJ, Sayre RT, Bauer WD. 2011. N-acyl homoserine lactone lactonase, AiiA, inactivation of quorum-sensing agonists produced by *Chlamydomonas reinhardtii* (Chlorophyta) and characterization of *aiiA*-transgenic algae. *J Phycol*. 47:1219–1227.
- Rao D, Webb JS, Holmström C, Case R, Low A, Steinberg P, Kjelleberg S. 2007. Low densities of epiphytic bacteria from the marine alga *Ulva australis* inhibit settlement of fouling organisms. *Appl Envir Microb*. 73:7844–7852.
- Riemann L, Azam F. 2002. Widespread N-acetyl-D-glucosamine uptake among pelagic marine bacteria and its ecological implications. *Appl Environ Microbiol*. 68:5554–5562.
- Ritchie KB. 2006. Regulation of microbial populations by coral surface mucus and mucus-associated bacteria. *Mar Ecol Prog Ser*. 322:1–14.
- Romero M, Diggle SP, Heeb S, Camara M, Otero A. 2008. Quorum sensing activity in *Anabaena* sp. PCC7120: identification of AiiC, a novel AHL-acylase. *FEMS Microbiol Lett*. 280:73–80.
- Santos OC, Pontes PV, Santos JF, Muricy G, Giambiagi-deMarval M, Laport MS. 2010. Isolation, characterization and phylogeny of sponge-associated bacteria with antimicrobial activities from Brazil. *Res Microbiol*. 161:604–612.
- Schultz MP, Bendick JA, Holm ER, Hertel WM. 2011. Economic impact of biofouling on a naval surface ship. *Biofouling*. 27:87–98.
- Seper A, Fengler VH, Roier S, Wolinski H, Kohlwein SD, Bishop AL, Camilli A, Reidl J, Schild S. 2011. Extracellular nucleases and extracellular DNA play important roles in *Vibrio cholerae* biofilm formation. *Mol Microbiol*. 82:1015–1037.
- Shao CL, Wang CY, Wei MY, Gu YC, She ZG, Qian PY, Lin YC. 2011. Aspergilones A and B, two benzylazaphilones with an unprecedented carbon skeleton from the gorgonian-derived *Aspergillus* sp. *Bioorgan Med Chem Lett*. 21:690–693.
- Shao CL, Wu HX, Wang CY, Liu QA, Xu Y, Wei MY, Qian PY, Gu YC, Zheng CJ, She ZG, Lin YC. 2011. Potent antifouling resorcylic acid lactones from the gorgonian-derived fungus *Cochliobolus lunatus*. *J Nat Prod*. 74:629–633.
- Shnit-Orland M, Kushmaro A. 2009. Coral mucus-associated bacteria: a possible first line of defense. *FEMS Microbiol Ecol*. 67:371–380.
- Silva-Aciades F, Riquelme C. 2008. Inhibition of attachment of some fouling diatoms and settlement of *Ulva lactuca* zoospores by film-forming bacterium and their extracellular products isolated from biofouled substrata in Northern Chile. *Electron J Biotech*. 11:12.

- Stahl LJ. 1995. Physiological ecology of cyanobacteria in microbial mats and other communities. *New Phytol.* 131:1–32.
- Steinberg PD, Rice SA, Campbell AH, McDougald D, Harder T. 2011. Interfaces between bacterial and eukaryotic “neuroecology”. *Integr Comp Biol.* 51:794–806.
- Tan LT, Goh BPL. 2009. Chemical ecology of marine cyanobacterial secondary metabolites: a minireview. *J Coastal Dev.* 13:1–8.
- Taylor MW, Schupp PJ, Baillie HJ, Charlton TS, de Nys R. 2004. Evidence for acyl homoserine lactone signal production in bacteria associated with marine sponges. *Appl Environ Microbiol.* 70:4387–4389.
- Teplitski M, Mathesius U, Rumbaugh KP. 2011. Perception and degradation of N-Acyl homoserine lactone quorum sensing signals by mammalian and plant cells. *Chem Rev.* 111:100–116.
- Thiyagarajan V. 2010. A review on the role of chemical cues in habitat selection by barnacles: new insights from larval proteomics. *J Exp Mar Biol Ecol.* 392:22–36.
- Tinh NTN, Dierckens K, Sorgeloos P, Bossier P. 2008. A review of the functionality of probiotics in the larviculture food chain. *Mar Biotechnol.* 10:1–12.
- Tommonaro G, Abbamondi GR, Iodice C, Tait K, De Rosa S. 2012. Diketopiperazines produced by the halophilic archaeon, *Haloterrigena hispanica*, activate AHL bioreporters. *Microb Ecol.* 63:490–495.
- Uroz S, Chhabra SR, Camara M, Williams P, Oger P, Dessaux Y. 2005. N-acylhomoserine lactone quorum-sensing molecules are modified and degraded by *Rhodococcus erythropolis* W2 by both amidolytic and novel oxidoreductase activities. *Microbiol-Sgm.* 151:3313–3322.
- Uroz S, D’Angelo-Picard C, Carlier A, Elasmri M, Sicot C, Petit A, Oger P, Faure D, Dessaux Y. 2003. Novel bacteria degrading N-acylhomoserine lactones and their use as quenchers of quorum-sensing-regulated functions of plant-pathogenic bacteria. *Microbiol-Sgm.* 149:1981–1989.
- Uroz S, Oger PM, Chapelle E, Adeline MT, Faure D, Dessaux Y. 2008. A *Rhodococcus* qsdA-encoded enzyme defines a novel class of large-spectrum quorum-quenching lactonases. *Appl Environ Microbiol.* 74:1357–1366.
- Videla HA, Herrera LK. 2005. Microbiologically influenced corrosion: looking to the future. *Int Microbiol.* 8:169–180.
- Volk RB, Furkert FH. 2006. Antialgal, antibacterial and antifungal activity of two metabolites produced and excreted by cyanobacteria during growth. *Microbiol Res.* 161:180–186.
- Volk RB, Girreser U, Al-Refai M, Laatsch H. 2009. Bromoanandolone, a novel antimicrobial exometabolite from the cyanobacterium *Anabaena constricta*. *Nat Prod Res.* 23:607–612.
- Vynne NJ, Mansson M, Nielsen KF, Gram L. 2011. Bioactivity, chemical profiling, and 16S rRNA-based phylogeny of *Pseudoalteromonas* strains collected on a global research cruise. *Mar Biotechnol.* 13:1062–1073.
- Wahl M. 1989. Marine epibiosis. I. Fouling and antifouling: some basic aspects. *Mar Ecol Progr Ser.* 58:175–189.
- Wahl M, Goecke F, Labes A, Dobretsov S, Weinberger F. 2012. The second skin: ecological role of epibiotic biofilms on marine organisms. *Front Microbiol* 292 (3), doi: 10.3389/fmicb.2012.00292. Available from: http://www.frontiersin.org/Aquatic_Microbiology/10.3389/fmicb.2012.00292/abstract
- Wang YN, Shao CL, Zheng CJ, Chen YY, Wang CY. 2011. Diveristy and antibacterial activities of fungi derived from the gorgonian *Echinogorgia rebekka* from the South China Sea. *Mar Drugs.* 9:1379–1390.
- Waters CM, Bassler BL. 2005. Quorum sensing: cell-to-cell communication in bacteria. *Annu Rev Cell Dev Biol.* 21:319–346.
- Webster NS, Cobb RE, Negri AP. 2008. Temperature thresholds for bacterial symbiosis with a sponge. *ISME J.* 2:830–842.
- Wiese J, Thiel V, Nagel K, Staufenberg T, Imhoff JF. 2009. Diversity of antibiotic-active bacteria associated with the brown alga *Laminaria saccharina* from the Baltic Sea. *Mar Biotechnol.* 11:287–300.
- Wilson GS, Raftos DA, Nair SV. 2011. Antimicrobial activity of surface attached marine bacteria in biofilms. *Microbiol Res.* 166:437–448.
- Xiong HR, Qi SH, Xu Y, Miao L, Qian PY. 2009. Antibiotic and antifouling compound production by the marine-derived fungus *Cladosporium* sp. F14 *J Hydro-Envir Res.* 2: 264–270.
- Xu Y, He HP, Schulz S, Liu X, Fusetani N, Xiong HR, Xiao X, Qian PY. 2010. Potent antifouling compounds produced by marine *Streptomyces*. *Bioresour Technol.* 101:1331–1336.
- Xu Y, Li HL, Li XC, Xian X, Qian PY. 2009. Inhibitory effects of a branched-chain fatty acid on larval settlement of the polychaete *Hydroides elegans*. *Mar Biotech.* 11:495–504.
- Yan LM, Boyd KG. 2002. Surface attachment induced production of antimicrobial compounds by marine epiphytic bacteria using modified roller bottle cultivation. *Mar Biotechnol.* 4:356–366.
- Yebrá DM, Kiil S, Dam-Johansen K. 2004. Antifouling technology- past, present and future steps towards efficient and environmentally friendly antifouling coatings. *Prog Org Coating.* 50:75–10.
- Yebrá DM, Kiil S, Weinell CE, Dam-Johansen K. 2006. Presence and effects of marine microbial biofilms on biocide-based antifouling paints. *Biofouling.* 22:33–41.
- Yee LH, Holmström C, Fuay ET, Lewin NC, Kjelleberg S, Steinberg PD. 2007. Inhibition of fouling by marine bacteria immobilised in kappa-carrageenan beads. *Biofouling.* 23:287–294.
- Zardus JD, Nedved B, Huang Y, Tran C, Hadfield MG. 2008. Microbial biofilms facilitate adhesion in biofouling invertebrates. *Biol Bull.* 214:91–98.
- Zargiel KA, Coogan JS, Swain GW. 2011. Diatom community structure on commercially available ship hull coatings. *Biofouling.* 27:955–963.
- Zhang XY, Bao J, Wang GH, He F, Xu XY, Qi SH. 2012a. Phylogenetic survey and antimicrobial activity of culturable microorganisms associated with the South China Sea black coral *Antipathes dichotoma*. *FEMS Microbiol Lett.* 336:122–130.
- Zhang XY, Bao J, Wang GH, He F, Xu XY, Qi SH. 2012b. Diversity and antimicrobial activity of culturable fungi isolated from six species of the South China Sea gorgonians. *Microb Ecol.* 64:617–627.